Dilan Karaguler

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Research Interests

Applied algebraic topology and topological data analysis; path homology and Mayer (N-nilpotent) homology of directed graphs and path complexes; persistent homology and persistent Laplacians; topology-informed graph machine learning and geometric deep learning, with applications to molecular, dynamical and engineering data.

Education

Michigan State University — East Lansing, MI, US
Ph.D. in Mathematics (Candidate) — August 2022 – May 2027 (Expected)
- Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

Middle East Technical University (METU) — Ankara, Türkiye
M.S. in Mathematics — September 2019 – September 2021
- Thesis: *A Survey on Multidimensional Persistence Theory*
- Advisor: Assoc. Prof. Semra Pamuk

Middle East Technical University (METU) — Ankara, Türkiye
B.S. in Mathematics, Honor Student — June 2014 – June 2019

Publications

D. Karaguler and G.-W. Wei. *Mayer Path Homology.* Foundations of Data Science, accepted (2026). doi:10.3934/fods.2026025, arXiv:2605.16525.

D. Karaguler and G.-W. Wei. *Persistent Mayer Path Homology and Laplacian.* Preprint (2026), to appear on arXiv.

D. Karaguler and G.-W. Wei. *Directed Topology for Protein–Nucleic Acid Flexibility.* In preparation (2026).

D. Karaguler. *A Survey on Multidimensional Persistence Theory.* M.S. Thesis, Middle East Technical University (2021).

Talks

Persistent Mayer Path Homology and Laplacian (upcoming) — November 2026
Applied Mathematics Student Seminar, Michigan State University — East Lansing, MI, US

Mayer Path Homology — April 2026
TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU — East Lansing, MI, US

Interaction Topology in TDA — October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU — East Lansing, MI, US

Mayer Topology in TDA — October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU — East Lansing, MI, US

From Data to Barcodes — April 2021
Graduate Student Seminar, Middle East Technical University — Ankara, Türkiye

Poster Presentations

Mayer Path Topology and Persistence — September 2025
SIAM Great Lakes Section Meeting, Illinois Institute of Technology — Chicago, IL, US

Mayer Path Topology and Persistence — March 2025
Mathematics of Data, Dynamics, and Life Sciences Conference, UC Irvine — Irvine, CA, US

Research Experience

Graduate Researcher August 2022 – Present
Michigan State University East Lansing, MI, US

- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Extended it to persistent Mayer path homology and the Mayer path Laplacian, with stability theorems and a molecular application (preprint, 2026).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

Research Mentor May 2023 – June 2023
Topology RTG, Michigan State University East Lansing, MI, US

- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

Industry Experience

Software Engineer Intern May 2026 – August 2026
Intel Corporation Oregon, US

- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

Graduate Technical Intern May 2025 – August 2025
Intel Corporation Oregon, US

- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

Projects

Persistent Mayer Homology with Dynamic Mode Decomposition April 2026
Python, NumPy, SciPy, GUDHI GitHub

- Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
- Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

Homology-Fed Neural Network Framework November 2024
Python, PyTorch, GUDHI, NumPy GitHub

- Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
- Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

Cellular Automata on Curved Surfaces March 2024
Python, NumPy, Matplotlib GitHub

- Simulated Conway's Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

Teaching Leadership & Mentoring

Lead Teaching Assistant August 2024 – Present
Department of Mathematics, Michigan State University East Lansing, MI, US
- **2026 – 2027:** Co-lead, with Eloy Moreno-Nadales, a yearlong workshop series for first-year graduate TAs: introductions, *Three Aspects of Teaching*, mock teaching with peer feedback, and a graduate student panel.
- **Fall 2025 (MTH 132):** Prepared recitation worksheets, gathered TA feedback for instructors, and substituted for recitation leaders.
- **Spring 2025 (MTH 133):** Ran TA meetings, prepared weekly worksheets, and set up Teams tools to track grading deadlines.
- **Fall 2024:** Ran ULA orientation and mock-teaching sessions, and observed ULAs and GTAs to give feedback.

Graduate Lead, Center for Instructional Mentoring (CIM) Summer 2024
Department of Mathematics, Michigan State University East Lansing, MI, US
- Mentored first-time graduate instructors through weekly check-ins and non-evaluative feedback.
- Created a teaching handbook from interviews with senior GTAs on each course's expectations.

Mentor, Academic Culture and English Skills (ACES) Program Summer 2024
Michigan State University East Lansing, MI, US
- Mentored incoming international graduate TAs on U.S. classroom culture and cross-cultural communication; organized social events.

Supervisor, Math Learning Center (MLC) Fall 2022 – Spring 2023, Fall 2024
Michigan State University (Teaching Exemption Fellowship, 2022 – 2023) East Lansing, MI, US
- Managed how tutors were distributed among students and arranged the room to match demand.

Student feedback: dilankaraguler.github.io/teaching/what-students-say

Courses Taught

MTH 314 — Matrix Algebra with Computational Applications Fall 2026
Instructor (1 section), Michigan State University East Lansing, MI, US
- Prepare and deliver lectures, facilitate group work, and hold help-room hours.
- Recorded per-question quiz scores to send each student personalized feedback on topics to review and problems to practice, and showed coordinators how quiz versions affected grades.

MTH 299 — Transitions Fall 2025
Teaching Assistant (2 sections), Michigan State University East Lansing, MI, US
- Co-taught with Ester Sgallova: weekly recitations, homework solutions, grading and help-room hours.

MTH 314 — Matrix Algebra with Computational Applications Spring 2025
Teaching Assistant (1 section), Michigan State University East Lansing, MI, US
- Graded homework and the final exam; ran quiz and exam reviews and help-room hours.

MTH 133 — Calculus II Fall 2024
Teaching Assistant (1 section) and pod leader, Michigan State University East Lansing, MI, US
- Led a recitation; as pod leader, ran recitation-leader meetings and wrote the weekly worksheets.

MTH 235 — Differential Equations Spring 2024
Teaching Assistant (3 sections), Michigan State University East Lansing, MI, US
- Wrote recitation notes with summaries and practice problems; graded quizzes and held help-room hours.

MTH 133 — Calculus II Fall 2023
Teaching Assistant (3 sections: 2 online, 1 in person), Michigan State University East Lansing, MI, US
- Prepared weekly lecture summaries; graded and proctored quizzes; held help-room hours and followed up with online students.

MTH 133 — Calculus II Summer 2023
Teaching Assistant (1 online asynchronous section), Michigan State University East Lansing, MI, US
- Contacted students weekly by email and tracked their course activity.

MTH 154, 153, 120, 119 — Calculus I and II January 2020 – June 2022
Graduate Teaching Assistant, Middle East Technical University Ankara, Türkiye
- MTH 154 (Spring 2022) and MTH 153 (Fall 2021) for Mathematics Students; MTH 120 (Spring 2021) and MTH 119 (Fall 2020) for Engineering Students.
- Led recitations and office hours; graded and proctored exams.

MTH 154 — Calculus II for Mathematics Students Spring 2019
Undergraduate Teaching Assistant (homework grader), Middle East Technical University Ankara, Türkiye
- Graded homework for more than 300 students.

MTH 153 — Calculus I for Mathematics Students Fall 2018
Undergraduate Teaching Assistant (homework grader), Middle East Technical University Ankara, Türkiye
- Graded homework for more than 300 students.

MTH 116 — Introduction to Abstract Algebra Spring 2017
Undergraduate Teaching Assistant (recitation instructor), Middle East Technical University Ankara, Türkiye
- Conducted recitations for 180 students with my own recitation materials.

Awards, Fellowships & Scholarships

Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University — Summer 2026
Excellence in Teaching Award, Department of Mathematics, Michigan State University — May 2025
Colleges Online Learning Academy (COLA) Fellowship, Michigan State University — Summer 2024
Teaching Exemption Fellowship, Michigan State University — 2022 – 2023
RONESANS Scholarship, RONESANS — 2018 – 2019
UNIGENC Scholarship, UNIGENC Association — 2015 – 2019
Merit Scholarship, Turkish Ministry of National Education — 2010 – 2014

Service, Leadership & Outreach

Graduate Representative, NatSci Student Advisory Council (SAC) — 2025 – 2026
College of Natural Science, Michigan State University — East Lansing, MI, US

Graduate Student Representative, Graduate Studies Committee — 2024 – 2025
Department of Mathematics, Michigan State University — East Lansing, MI, US

- Represented mathematics graduate students on the committee and co-planned Responsible Conduct of Research (RCR) workshops.
- Hosted town halls, met prospective students, and directed students with concerns to department resources.

Member, First-Generation/Low-Income (FGLI) Group — September 2024 – May 2025
Department of Mathematics, Michigan State University — East Lansing, MI, US

- Took part in community gatherings and social events with first-generation and low-income undergraduates.

Organizing Committee Member — April 2024
Graduate Student Topology and Geometry Conference, Michigan State University — East Lansing, MI, US

- Handled speaker communication, scheduling and room assignments; moderated sessions.

President / Board Member — August 2022 – August 2025
Association for Women in Mathematics (AWM), MSU Chapter — East Lansing, MI, US

- Organized seminars, career workshops, academic-life sessions and social events for women in mathematics.

President — February 2015 – December 2018
METU Math Club — Ankara, Türkiye

- Organized seminars, workshops and reading groups; prepared first-year students to lead abstract algebra and number theory discussions.

Outreach & Volunteering

Proctor, Math Kangaroo Team Competition, Department of Mathematics, MSU (upcoming) — November 2026
Volunteer, Math Night, Marble Elementary School, East Lansing — November 2026
Volunteer, MSU Student Food Bank, Michigan State University — Spring 2024

Technical Skills

Programming: Python, R, SQL
Methods: Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation
Libraries: PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM
Tools: Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, LaTeX
Domain Knowledge: Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis

References

Available upon request.